

# ESTIMATING TMF COSTS

This study demonstrates one way to estimate the costs associated with managing a paper–based Trial Master File (TMF). Having those expenses in hand enables the reader to present a compelling picture of the potential savings that result from eliminating the need to print, ship, sign, and file key clinical trial regulatory documents.

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# Improving Clinical Trial ROI with Automation

## Executive Summary

Clinical trials are burdened with mountains of paper and manual processes. The labor costs involved in the manual processes of collecting, completing, classifying and reviewing investigator documents are significant and are a significant cost driver in clinical trials. However, leading BioPharma organizations are increasingly using collaborative cloud computing and portal based application solutions to create, digitally sign and complete documents in the cloud through automated processes called workflows. Workflow task automation helps to eliminate paper at the point of origination, reducing or eliminating the need for labor intensive processes like document printing and filing, scanning, manual tagging and classification. With automated workflows, clinical trial operations can realize significant cost and time savings while realizing higher levels of clinical trial data management quality, process compliance, reporting and auditability.

SureClinical offers next generation collaborative cloud, portal-based applications that can be used by all stakeholders in a clinical trial project – study sponsors, clinical research organizations, investigators, auditors and data providers. SureClinical is the only clinical trial eTMF applications provider to offer the ability to organizations to create, digitally sign and complete documents entirely in the cloud with FDA and EC compliant Adobe® digital ID's, eliminating paper at the point of origination for clinical trial operations. SureClinical's cloud applications are accessible across web and mobile devices, providing anytime, anywhere access to global teams to complete tasks on time and on schedule through a simple and easy to use wizard based web portal. By empowering the entire clinical stakeholder team with access to cloud based portals and task completion with automated workflows, manual processes can be reduced or eliminated. As a result of task automation, enhanced clinical trial project management, comprehensive task tracking, milestone management and oversight are possible, accelerating clinical trial project completion.

> According to AIIM, the global community of information professionals, "60% of users have seen ROI on their paper-free projects within 12 months, an impressive 77% within 18 months.
>
> Faster response and higher productivity are seen as the biggest benefits, along with improved remote and mobile availability."
>
> *—2014 AIIM Market Intelligence Report, 'Paper Wars'*

## COLLABORATIVE CLOUD PORTALS ENHANCE CLINICAL TRIAL OPERATIONS

In the fast-paced BioPharma business environment, collaborative cloud computing is gaining traction, providing enterprises with a means to improve operational efficiency and accelerate time to market. Collaborative cloud computing is at the center of a movement in the software industry to build functionally rich and vertically-focused solutions that can extend the power of enterprise computing beyond the firewall to include external business partners, in what is commonly referred to as Extraprise collaborative cloud computing.

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Collaborative processes play a role in almost every part of your business – from clinical trial research and development, clinical trial study startup to product marketing. The applications and technology infrastructure you use are essential to leverage and integrate the work of your entire internal and external clinical trial team, including sponsors, CROs, investigators, academic research organizations and other clinical trial partners.

Although cloud computing applications are only beginning to be adopted by life sciences companies, industry analysts predict that cloud-based, collaborative clinical trial portals will be used extensively to accelerate clinical trial development, reduce clinical trial study costs, enhance clinical trial operations efficiency and improve quality. [1] Clinical trial portals provide a centralized point of collaboration to complete tasks online. By utilizing new technologies such as integrated digital signing, portal users can collaboratively create, review and digitally sign documents entirely online, eliminating the cost and waste of manually printing, scanning, uploading, tagging and classifying documents.

SureClinical sees collaborative cloud applications, clinical trial portals and automated workflows as a means of ensuring process consistency, efficiency and enhanced regulatory compliance. Automated workflows with integrated digital signing technologies enable portal-based team collaboration and document completion across web, mobile and desktop platforms for centralized management, tracking, reporting, auditing and quality management. In addition to the cost and time savings of automated processes over manual processes, automated workflows and automated task completion can enhance your ability to comply with global FDA, HIPAA, EU and other regulatory agency requirements through enhanced process monitoring and oversight.

## REQUIREMENTS FOR COLLABORATIVE CLOUD PORTALS WITH TASK AUTOMATION

Collaborative cloud portals are used in clinical trial regulatory document management to connect people, organizations, and clinical trial applications and study content in a standards-based, compliant, secure environment. Instead of manually processing clinical trial documents using time-consuming tasks such as printing, signing, and scanning, today's cloud-based applications simplify and automate tasks by enabling users and clinical trial partners to complete documents entirely in the cloud using task automation, otherwise known as workflows.

> Two thirds of users recognize the importance of mobile devices for content access and data capture.
>
> *—2014 AIIM Market Intelligence Report, 'Paper Wars'*

Companies are embracing collaborative cloud computing solutions that enable secure information sharing and FDA and EU agency compliant digital signatures on documents. This facilitates document completion beyond the corporate firewall with business partners, fundamentally transforming the way that clinical trials are conducted. Best-in-class collaborative cloud applications allow users from multiple organizations to collaboratively review, digitally sign and complete documents through a centralized, standards-based, collaborative cloud application that is secure and compliant. The shared cloud portal approach generates economies of scale that reduces costs, accelerates new drug delivery times and facilitates global business organizational agility and flexibility.

To realize these benefits, BioPharma organizations are deploying collaborative cloud solutions that connect global clinical trial teams together seamlessly within a compliant, standards-based environment.

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Collaborative cloud portals need to be able to provide an enterprise-level solution that can do the following:

- Quickly connect teams of sponsors, CROs and investigators in a secure portal with a role-based experience, available on mobile phones, tablets and web devices
- Eliminate paper-based processes with digital signing and automated document workflows to better manage data
- Track essential document milestone completion with reporting and audit trails for compliance and quality
- Improve collaboration and document exchange capabilities to accelerate the flow of information between stakeholders
- Support standards-based technologies to facilitate interoperability with existing systems

## CLINICAL TRIAL TASK AUTOMATION WITH WORKFLOWS

A key system requirement for any organization who wants to automate clinical trial task completion would include integrated support for completing tasks through a secure portal using workflow automation. A workflow is simply a series of tasks that are part of a business process. As an example, sending a document for someone to sign is an example of a basic workflow process. Workflows can be used to consistently manage common clinical trial processes within an organization. Instead of manually performing tasks such as document signing, review and classification, organizations specify a set of tasks, documents, instructions and the persons to complete a process. The eClinical application software then completes the task automatically.

Workflows can streamline the cost and time required to coordinate common clinical trial processes, such as document review, approval and digital signing, by managing and tracking the human tasks involved with these processes. For example, with SureClinical eTMF, you can create a workflow from our eClinical workflow library that routes a protocol signature document to a group of principal investigators for review and digital signature.

The study administrator starts this workflow by selecting the protocol signature document or form from the eTMF system, then specifies the list of principal investigators (workflow recipients) who need to review and sign the document. The system creates and launches the workflow and sends email task notification alerts to the participants with task instructions and a link to the document to be reviewed and digitally signed. While the workflow is in progress, the workflow initiator (in this case, the study administrator) or the workflow participants can check the Workflow dashboard page to see which participants have completed their workflow tasks. When the workflow participants complete their workflow tasks, the workflow ends, and the workflow owner is automatically notified that the workflow has completed.

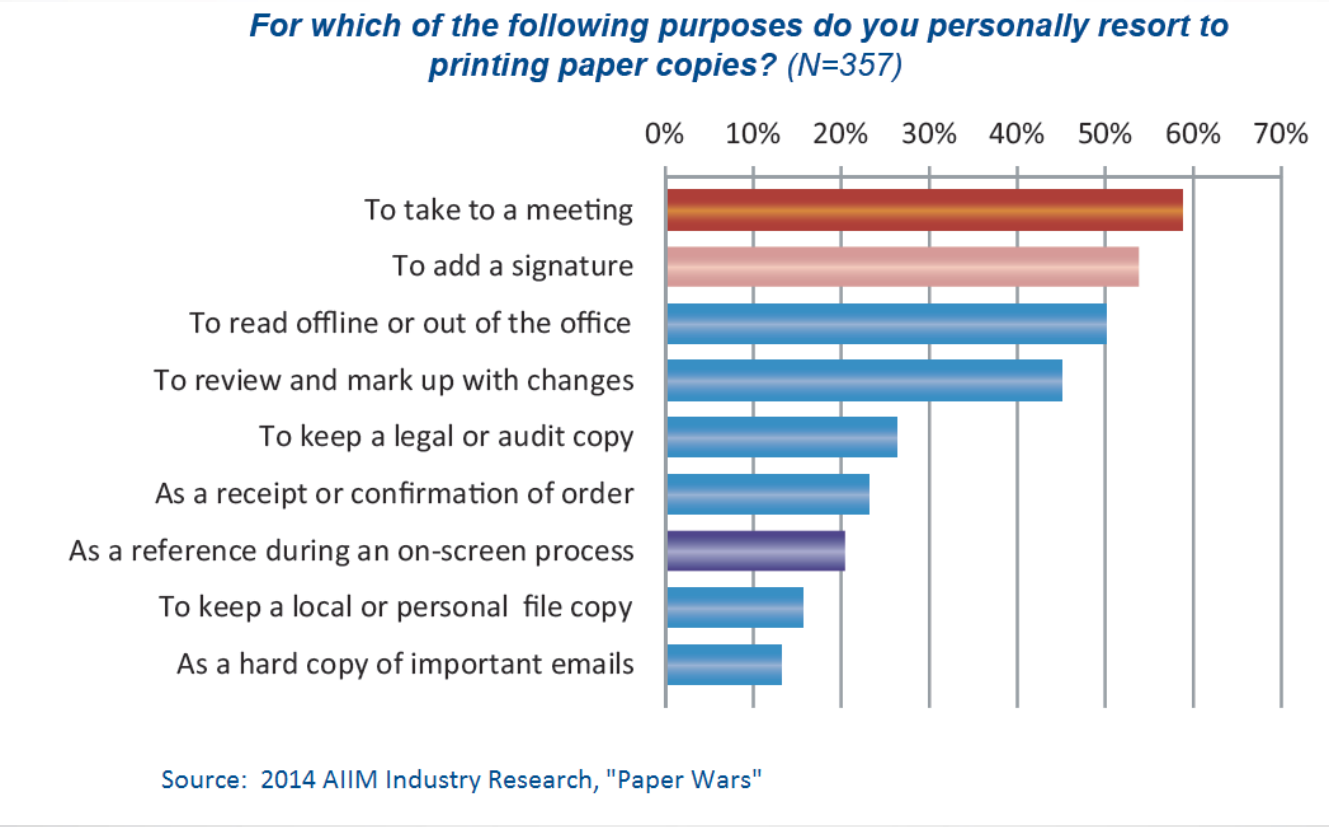
Contact SureClinical to have one of our consultants perform a free estimate of your study's current paper-based costs compared to your estimated eTMF costs using eClinical workflows and our FDA/EC compliant Adobe digital signing.

SureClinical provides the only eClinical collaborative cloud portal, applications and platform that has integrated workflows with integrated Adobe digital signing. SureClinical provides a library of customizable eClinical workflows, allowing organizations the flexibility to create, execute and completed organization specific eClinical workflows. Detailed audit trails, reporting and dashboards provide visibility and control over workflow processes and tasks.

## CLINICAL TRIAL PROCESS AUTOMATION – BEST PRACTICES

As clinical trial organizations contemplate implementation of systems and portals that can automate clinical trial tasks such as essential document collection, it's important to first benchmark and summarize existing manual paper-based clinical trial processes. After benchmarking existing processes and outlining the key objectives, documents and resources, persons, steps and decision points that comprise existing processes, organizations can create an optimized series of automated task steps that map automate tasks to existing manual tasks. These automated task steps are then completed through an automated workflow process.

Key factors to assess when deploying a workflow include ease of use, ease of access by stakeholders with support from popular mobile and web devices, ease of deployment, ease of administration, security, collaboration and regulatory compliance.




To help illustrate how organizations can recognize enhanced return on investment for its clinical trial operations, on the next few pages we'll outline a few examples of best practices to automated common clinical trial document collection tasks in clinical trials.

In all cases, labor is the single biggest contributor to the total cost of the clinical study. In the examples below, only the costs for the various labor categories are calculated. The reader can then determine the percent of labor that will be saved as a result of adopting an eTMF. Labor categories that are relevant to paper-handling costs include document handling, scanning, document tagging and filing, and reporting.

## Document Handling

Document handling includes management, preparation, printing, and shipping of documents. Typically, clinical operations can expect one full-time equivalent (FTE) to spend about four to five hours per month in document handling. Many of these individuals are TMF document consultants. For the purpose of these calculations, SureClinical assumed a $95 per hour rate for this category. For a study comprising a few hundred sites, the clinical operations manager can anticipate spending at least $1M per year to manage documents.

## Scanning

Not only is the cost of capital investment in scanners sizable, but the labor to maintain and process scanned documents is even greater! Scanning includes the labor to scan signed documents into a digital archive or to scan documents for email distribution and digital signing. Assuming a $50 per hour labor rate, 4 hours per month per site, and the same study characteristics as above, the total budget will run approximately $480,000 per year for scanning labor.

## Document Tagging and Filing

Document tagging and filing includes assigning the correct TMF naming conventions to the files, and then physically filing essential regulatory documents in fireproof cabinets. Given the same rate of pay as scanning, and assuming that clinical operations personnel spend just an hour per month filing at each site, the cost of document tagging and filing can be as much as $120,000 per year.

## Report Preparation

The final key area of administrative labor includes the generation of reports to show documentation completion status, to resolve discrepancies, and to provide other key process indicators (KPIs) necessary for completion of a successful study. For the purpose of this calculation, we assumed a rate of $95 per hour for a seasoned auditor to generate and manage audit reports. At that rate, clinical operations can expect to spend as much as $450,000 per year in reporting costs.

## Total Direct Costs - Paper TMF

So what is the total annual spend to support a paper-based clinical trial? Incredibly, a large global study can easily spend nearly $3.5M per year administering paper processes. More amazing is that converting to an eTMF that automates the totality of the regulatory content management can eliminate the vast majority of that amount. For any cost-conscious executive, returning three and a half million dollars to the research and development budget is the obvious choice.

**Total Direct Costs - Assume 200 Global Sites:**

- *Document Handling, Overnight Mail*
- *Scanning, Printing, Storage, Other*
- *Labor - Tagging and Classification*
- *Report Creation*
- *Audit travel expenses*

**$3.5M / year**

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## SureClinical Certified Cloud™

The SureClinical's Certified Cloud™ is an FDA Part 11 and HIPAA compliance validated applications suite that automate the acquisition, classification, signing, auditing, and archiving of clinical study regulatory documents. An efficient Software as a Service (SaaS) solution, SureClinical can be activated in minutes. The SureClinical solution generates an immediate ROI by eliminating paper-based processes, saving valuable time, energy, and dollars.

## Free eTMF Return on Investment Consultation

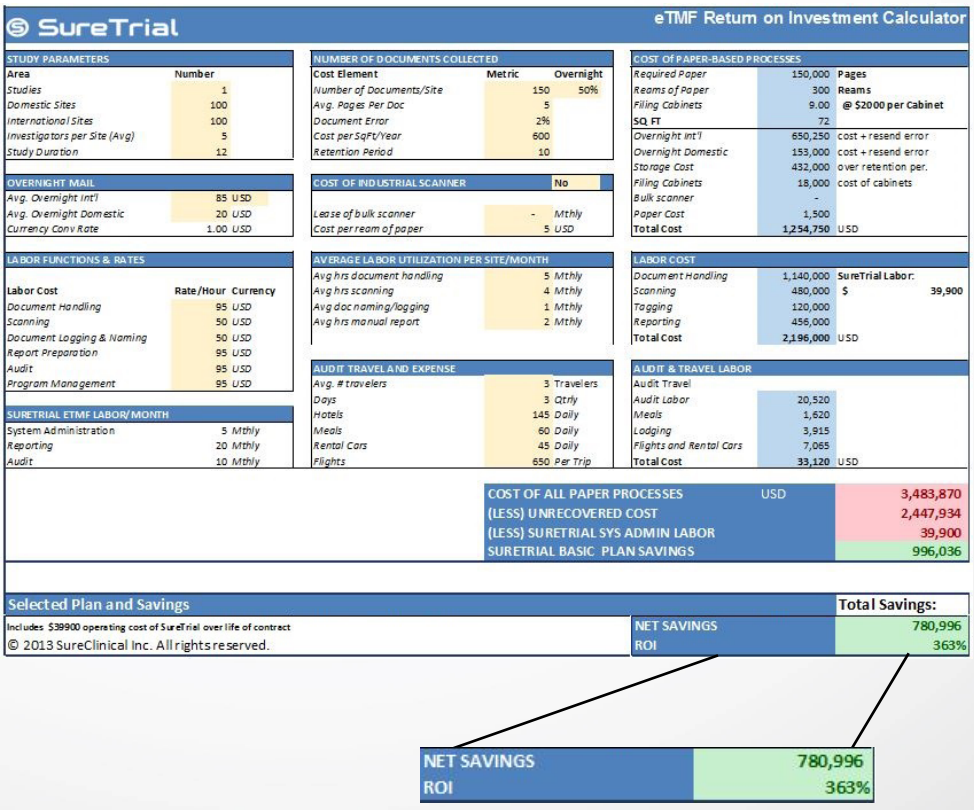
Our team of experts is prepared to work with you to estimate your current TMF costs and the estimated savings and total return on investment using the SureClinical eTMF Certified Cloud. Take advantage of this valu-able consultation by contacting us today!

Contact SureClinical for an estimate of your study's paper-based costs. Learn how you can increase efficiency and lower study costs with an eTMF.

Call (916) 265-2000 or email:

info@sureclinical.com

## SureClinical eTMF ROI Calculator




—Estimate your paper TMF costs

—Estimate Net Savings with SureClinical eTMF Cloud


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